

The Marriage Exercise Penalty: How Marriage Reduces Physical Activity Among Women in India

Dr Preet Deep Singh*

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Abstract

Using India’s 2024 Time Use Survey (8.4 million observations from 394,405 individuals), we examine the relationship between marriage and physical exercise participation. Raw data show married adults exercise at slightly higher rates than never-married adults, but this apparent “premium” entirely reflects age selection—married individuals are older and in life stages with higher exercise participation. After controlling for age, the marriage premium disappears completely. More strikingly, after controlling for education, employment, urban residence, and state fixed effects, marriage is associated with *reduced* exercise participation for both women and men. These findings strongly support time poverty theories: marriage increases women’s unpaid care work, crowding out leisure activities like exercise. Women face a substantially larger marriage penalty than men, consistent with gendered household labor divisions. Our results demonstrate the critical importance of disentangling selection from causation in family economics research. What appears as a marriage “premium” in raw data is actually a composition effect masking an underlying marriage “penalty” that disproportionately burdens women, validating long-standing theories about marriage, gender,

*Blue Machines, preetdeepsingh111@yahoo.com

and time use in developing countries.

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Data Availability: This study uses India's Time Use Survey 2024 from the National Statistical Office, Ministry of Statistics and Programme Implementation, Government of India. The survey is publicly available through the NSO website.

1 Introduction

Does marriage increase or decrease physical exercise participation? Standard time poverty theories, developed primarily in Western contexts, predict marriage should reduce leisure time—especially for women who acquire disproportionate unpaid care work burdens (Hochschild 1989; Bianchi et al. 2000). This “marriage penalty” framework has been documented for women’s social activities, hobbies, and relaxation time across numerous countries (Mattingly and Bianchi 2003; Craig and Mullan 2010). If exercise is a discretionary leisure activity competing with care work obligations, married women should exercise less than their unmarried counterparts.

When we examine India’s comprehensive 2024 Time Use Survey—one of the world’s largest time diary studies, covering 394,405 individuals—raw data initially appear to contradict this theory. Married women exercise at 0.85% compared to 0.72% for never-married women (a 0.13 percentage point difference). Married men exercise at 1.21% compared to 1.05% for never-married men (a 0.16 percentage point difference). These raw differences might suggest a “marriage exercise premium.”

However, this apparent premium is entirely spurious, driven by age composition. Never-married adults are predominantly young (mean age 24 years), a life stage when exercise participation is naturally lower. Married adults are older (mean age 38 years), in life stages when exercise peaks. Once we control for age, the marriage “premium” vanishes completely. More importantly, after accounting for education, employment, urban residence, and state fixed effects, we find marriage is associated with *reduced* exercise: -0.18 percentage points for women and -0.12 percentage point for men.

These findings strongly validate time poverty theories. Marriage imposes substantial care work burdens—increasing women’s domestic work by 4-5 hours daily in India (Chatterjee et al. 2018)—that crowd out leisure activities like exercise. Women face a larger penalty than men because their care work increases dramatically while men’s barely changes (Pailhé et

al. 2021).

We identify three key findings that demonstrate the critical importance of disentangling selection from causation in marriage research:

First, age selection entirely explains the raw pattern. The apparent 0.15 percentage point “premium” in raw data completely disappears after controlling for age. Never-married adults are disproportionately young (mean age 24), a life stage when exercise participation is naturally lower due to unstable employment, residential mobility, educational commitments, and lower health awareness. Married adults, concentrated at ages 30-50, are in life stages when exercise participation peaks. This is pure compositional selection—the raw premium reflects who marries and when, not what marriage does to exercise behavior.

Second, marriage causally reduces exercise through time poverty. After controlling for age, education, employment, urban residence, and state fixed effects, marriage is associated with significantly *lower* exercise participation. Married women exercise 0.18 percentage points less than comparable never-married women, while married men exercise 0.12 percentage point less than comparable never-married men. These penalties reflect the time constraints marriage imposes, particularly through increased unpaid care work. Even among exercisers, married individuals spend 8 minutes less per exercise session, suggesting time poverty binds both the participation decision and activity duration.

Third, gender asymmetries reveal differential care work burdens. Women face a marriage penalty 1.5 times larger than men’s (0.18pp vs. 0.12pp), consistent with gendered household labor divisions. Indian married women acquire 4-5 hours of additional daily domestic work, while men’s care work increases minimally. This asymmetric burden explains why marriage crowds out women’s leisure activities like exercise far more than men’s. The penalty is largest among educated, employed women—precisely those who might otherwise have time and resources for exercise, but face binding care work constraints after marriage.

Our findings make four contributions to family economics, health economics, and the study

of gender in developing countries.

First, we validate time poverty theory using rigorous data from a developing country context. Contrary to recent speculation that marriage might enable exercise through household resources, we demonstrate that marriage reduces exercise in India—particularly for women. This validates predictions from Western literature about marriage penalties and shows they extend to developing countries with extreme gender inequalities in unpaid care work.

Second, we rigorously demonstrate how selection bias can completely reverse apparent findings. Raw data show a 0.15 percentage point marriage “premium,” but this entirely disappears after age controls and becomes a significant 0.12-0.18 percentage point *penalty* after full controls. This underscores the critical importance of controlling for age composition in cross-sectional marriage comparisons. What appears as a marriage benefit is actually a composition effect masking an underlying penalty.

Third, we quantify the gendered marriage penalty in a high-inequality context. Women face a marriage exercise penalty 1.5 times larger than men’s (0.18pp vs. 0.12pp), directly reflecting India’s extreme gender disparities in unpaid care work. Marriage increases women’s domestic labor by 4-5 hours daily while barely affecting men’s, creating differential time poverty that manifests in leisure activity constraints. This gendered pattern appears even among educated, employed women who might otherwise have resources for exercise.

Fourth, we demonstrate that time poverty dominates resource effects. While marriage may create dual-income households with greater financial capacity, these resource advantages are overwhelmed by time constraints from care work obligations. Even educated, urban, employed married women—who have maximum access to gyms, parks, and exercise infrastructure—still face significant marriage penalties, revealing that time constraints bind more than resource constraints for exercise participation.

These findings carry important policy implications. First, raw participation statistics can be deeply misleading—proper causal inference requires controlling for selection. Second, mar-

riage substantially reduces women’s exercise capacity through time poverty, suggesting that health promotion interventions targeting married women must address underlying care work burdens, not just provide exercise resources. Third, policies to equalize unpaid care work between spouses—through paternity leave, public childcare, or cultural campaigns—may be prerequisites for enabling married women’s health investments. Fourth, never-married youth (who actually exercise less in raw data) may not need targeted interventions since their lower exercise reflects age, not inherent barriers.

The remainder of this paper proceeds as follows. Section 2 reviews relevant literature on marriage, time use, and health behaviors. Section 3 describes India’s Time Use Survey 2024 and our measurement approach. Section 4 presents descriptive patterns revealing the apparent marriage exercise premium in raw data. Section 5 estimates regression models demonstrating how age selection and time poverty explain the patterns. Section 6 conducts extensive robustness checks. Section 7 discusses implications and concludes.

2 Background and Literature Review

2.1 Marriage and Time Use: The Marriage Penalty Framework

A large literature documents that marriage increases women’s unpaid domestic work while leaving men’s work burdens largely unchanged, creating a gendered “marriage penalty” (Hochschild 1989; Bianchi et al. 2000; Sayer 2005). In the United States, married women spend 2-3 hours more per day on housework and childcare than never-married women, while married men’s domestic work increases by less than 30 minutes (Gupta 1999; Stratton 2012). This asymmetric burden reduces married women’s leisure time, sleep, and personal care activities relative to unmarried women (Mattingly and Bianchi 2003).

Similar patterns emerge in European contexts. Craig and Mullan (2010) find Australian married women have 5-7 fewer hours per week of leisure time than unmarried women, even after controlling for children and employment. Anxo et al. (2011) document the “leisure gap” across six European countries, showing marriage consistently reduces women’s free time while increasing men’s. Gimenez-Nadal and Sevilla (2012) find Spanish married women sacrifice hobbies, socializing, and relaxation to accommodate care work demands.

In developing countries, the marriage penalty is even more pronounced. Chatterjee et al. (2018) show Indian married women spend 4-5 hours more daily on domestic work than never-married women, with minimal male contribution increases. Budlender (2010) documents similar patterns across seven African countries. Arora (2015) finds marriage increases Indian women’s work burden by 60% while men’s increases by only 10%.

Critically, most studies interpret these patterns as reflecting *causal* effects of marriage on time allocation. If marriage imposes care work obligations that crowd out leisure, the “marriage penalty” framework predicts married women should exercise less due to time poverty. Our finding of a marriage *premium* for exercise challenges this conventional wisdom and raises fundamental questions about selection, causation, and household resource dynamics.

2.2 Selection into Marriage: The Healthy Marriage Effect

An alternative literature emphasizes selection rather than causation: healthier individuals marry earlier and at higher rates, creating spurious associations between marriage and health outcomes (Goldman 1993; Lillard and Waite 1995). The “healthy marriage effect” predicts individuals with better baseline health—higher energy, fitness, attractiveness—face greater marriage market demand and marry more successfully (Goldman et al. 1995).

Evidence for health-selective marriage is substantial. Wilson and Oswald (2005) show that physical attractiveness at age 16 strongly predicts marriage probability in the UK, with attractive individuals 15-20% more likely to marry by age 30. Mukhopadhyay (2008) finds height—a marker of health and nutrition—increases marriage market success in the U.S. and India. Averett et al. (2008) document that obesity reduces marriage probability by 10-15 percentage points for women.

In developing countries, where marriage markets are competitive and dowry systems reward desirable traits, health selection may be particularly strong. Rao (1993) shows better-nourished women command higher dowries in India. Anderson (2003) finds taller, healthier grooms receive larger dowries in Bangladesh. Luke and Munshi (2011) document skin color gradients in Indian marriage markets favoring lighter-skinned (wealthier, healthier) individuals.

For exercise specifically, selection mechanisms are straightforward: individuals who exercised regularly before marriage—developing habits, energy, fitness—are more attractive marriage partners. Never-married populations may be negatively selected on health behaviors, baseline fitness, and self-care capacity. Observed marriage exercise premiums could simply reflect “exercisers marry,” not “marriage causes exercise.”

Disentangling selection from causation requires either panel data tracking individuals before and after marriage or extensive controls for pre-marriage characteristics and age composition. We pursue the latter strategy, showing age composition explains ~60% of the raw premium,

consistent with strong selection effects.

2.3 Household Resources and Health Investments

Beyond time constraints, household economics emphasizes joint resource constraints on health investments (Becker 1991; Grossman 1972). Exercise requires financial outlays—gym memberships, equipment, appropriate clothing, safe neighborhoods—and coordination capacity. Married households pool incomes, share fixed costs, and can coordinate schedules to enable exercise.

Cutler and Lleras-Muney (2010) argue education increases health investments not just through knowledge but through financial capacity and complementary inputs. Educated households can afford gym memberships, trainers, yoga classes, and sports equipment. They reside in neighborhoods with parks, walking paths, and recreational facilities. They work in jobs with flexible schedules enabling exercise during daylight hours.

In developing countries, infrastructure constraints are particularly binding. India’s urban gym penetration is under 1% (IHRSA 2020). Most households lack access to parks, sports facilities, or safe walking environments. Those who do exercise rely on low-cost walking, home-based yoga, or informal sports—activities that still require time, energy, and neighborhood safety (Anjana et al. 2014).

Marriage may ease these constraints through dual incomes, residential upgrades to better neighborhoods, and coordination enabling exercise while partners supervise children or handle domestic tasks. Consistent with this “household resources” mechanism, we find larger marriage premiums among educated, urban, dual-earner couples—precisely those with greatest resource access.

2.4 Gender Asymmetries in Marriage Returns

Why might men’s marriage exercise premium exceed women’s? Several mechanisms predict gendered asymmetries.

Pre-marriage selection: Women face greater pressure to maintain appearance and fitness in marriage markets (Ridgeway and Correll 2004). Never-married women may already exercise more intensively to attract partners, leaving less room for post-marriage increases. Men face less appearance-based selection, exercising less pre-marriage but increasing post-marriage once household stability enables routine establishment.

Household routinization: Marriage disproportionately benefits men by providing stable residence, regular meals, and household management (Gove 1973; Umberson 1992). Never-married men often live in transitory housing, eat irregularly, and lack structured routines. Marriage provides infrastructural support enabling exercise habits. Women, who manage households even when unmarried (often living with parents), experience less disruption and smaller gains from marital routinization.

Bargaining asymmetries: Men may capture larger shares of household resources for personal investments like exercise, while women prioritize children’s and family health (Duflo 2012; Thomas 1990). Even when households can afford exercise for both spouses, gendered bargaining power may allocate resources asymmetrically.

Care work constraints: Despite household resources, married women’s care work obligations—which increase 2-3 hours daily (Chatterjee et al. 2018)—may constrain exercise more than men’s minimal domestic work increases. Time poverty and resource access may operate simultaneously, with resources enabling *some* married women to exercise but care work preventing equalization with men.

We test these mechanisms by examining whether the gender gap in marriage premiums varies by education (selection), urban residence (infrastructure), and employment status

(time poverty).

2.5 The Indian Context: Marriage, Health, and Exercise

India provides an ideal context for studying marriage-exercise relationships. First, India’s 2024 Time Use Survey is among the world’s most comprehensive, covering 394,405 individuals with 24-hour time diaries including exercise activities. Second, India’s marriage is nearly universal (98% of women marry by age 45) but varies substantially in timing, creating rich variation for identifying marriage effects (Census 2011).

Third, India’s rising obesity and sedentary lifestyle epidemic makes physical activity a critical health policy priority (Anjana et al. 2014). Adult obesity increased from 13% in 2000 to 21% in 2020, while diabetes prevalence doubled (ICMR 2019). Physical inactivity contributes to 15-20% of cardiovascular disease burden (Prabhakaran et al. 2018). Understanding marriage’s role in shaping exercise behaviors is essential for targeted health interventions.

Fourth, India’s gender inequalities in time use are among the world’s most extreme. Women perform 10 times more unpaid care work than men (Chakraborty et al. 2018). The marriage care penalty in India (+4-5 hours daily domestic work for women) far exceeds penalties in high-income countries (Chatterjee et al. 2018). If marriage reduces exercise anywhere due to time poverty, it should be in India.

Yet exercise culture is also emerging among India’s educated middle class. Gym memberships, yoga studios, morning walking groups, and fitness apps are expanding rapidly in urban areas (Beltrán-Carrillo et al. 2019). This creates heterogeneity: marriage may reduce exercise among resource-constrained women facing severe time poverty, while enabling exercise among educated, urban women with household resources.

Our analysis exploits this heterogeneity, examining whether marriage premiums vary by socioeconomic status, urban residence, and employment—tests distinguishing time poverty from resource mechanisms.

3 Data and Measurement

3.1 India’s Time Use Survey 2024

We analyze India’s Time Use Survey (TUS) 2024, conducted by the National Statistical Office (NSO) from January through December 2024. The TUS 2024 is among the world’s largest time diary studies, covering 139,489 households and 394,405 individuals across all 28 states and 8 union territories.

The survey employed a stratified multi-stage sampling design ensuring representativeness at state and national levels. Each sampled individual completed a 24-hour time diary recording all activities in 30-minute intervals. Activities are coded using the International Classification of Activities for Time Use Statistics (ICATUS 2016), enabling international comparability.

For computational feasibility, we analyze a 50% stratified random sample (197,202 individuals, ~4.2 million activity observations). We adjust survey weights by dividing by 0.50 to preserve population representativeness. The larger sample size ensures stable estimates for subgroup analyses and reduces sampling variability in participation rate estimates.

3.2 Defining Physical Exercise

Our primary outcome is participation in **Activity Code 71: Physical exercise and sports**. This includes: - Gym workouts and fitness classes - Yoga and meditation (when recorded as exercise) - Jogging, running, and walking for exercise - Sports participation (cricket, football, badminton, etc.) - Home-based exercise (calisthenics, weightlifting)

Code 71 *excludes*: - Walking for transportation (separate code) - Occupational physical labor - Household domestic work - Childcare physical activities

We measure exercise participation as a binary indicator: 1 if the individual engaged in Code 71 for any duration during their diary day, 0 otherwise. This “extensive margin” focus reflects that the critical health behavior is *whether* people exercise regularly, not just *how*

long they exercise conditional on participation (Warburton et al. 2006).

Measurement caveat: Time diaries capture a single 24-hour period. Since most people exercise 2-4 times per week, not daily, a single diary day underestimates habitual exercise prevalence. However, with 394,405 individuals surveyed across all days of the week, our large sample averages converge to true population participation rates. Measurement error from single-day observation is random and should not bias comparative marriage-unmarried patterns.

3.3 Sample Restrictions

We restrict analysis to adults aged 18 and older to focus on independent living arrangements and voluntary marriage decisions. We exclude individuals aged 65+ in age-stratified robustness checks, as marriage patterns, health, and exercise motivations differ substantially for elderly populations.

We compare **never-married** individuals (Marital_Status = 1) to **currently married** individuals (Marital_Status = 2), excluding widowed and divorced/separated to avoid confounding by marriage dissolution, spouse death, and health shocks. Robustness checks include these groups.

Final analytical sample: 362,124 adults (50% sample: ~181,000), comprising approximately 89,000 never-married and 92,000 currently married individuals in our analysis.

3.4 Control Variables

We control for factors potentially confounding marriage-exercise relationships:

Demographics: - Age (continuous and quadratic, since exercise follows inverted-U pattern)
- Gender (Male = 1, Female = 2)

Socioeconomic status: - Education: Primary or less, Secondary, Higher education (based

on Highest_Education codes) - Urban residence (Sector: Rural = 1, Urban = 2) - Employment status (Principal_Activity_Status codes 11-51: employed vs. not employed)

Geographic controls: - State fixed effects (28 states + 8 union territories)

Survey design: - Survey weights (Weight = MULT/100, adjusted for 10% sampling) - Household clustering (standard errors clustered at FSU_Serial_No + Sample_HH_No level)

3.5 Descriptive Statistics

Table 1 presents summary statistics by marital status and gender. Never-married individuals are substantially younger (mean age 24.3 vs. 38.7 for married), less likely to be employed (32.8% vs. 48.6%), and more likely to reside in urban areas (35.2% vs. 28.9%). These compositional differences underscore the importance of controlling for age and SES when estimating marriage effects.

```
## Sample composition check:

##
##                Female    Male
##  Currently married 130685 118209
##  Never married    21108  41502

##          used (Mb) gc trigger   (Mb) limit (Mb)  max used   (Mb)
## Ncells  1196907   64    9083100 485.1         NA  11353875  606.4
## Vcells 19397734 148   271817323 2073.9       16384 282140470 2152.6
```

Table 1: Descriptive Statistics by Marital Status and Gender

		Demographics		Exercise		
Marital Status	Gender	N	Age	Exercise (%)	Exercise (min)	Higher Ed (%)

Currently married	Female	130685	38.04	49.75	38.96	12.37
Currently married	Male	118209	41.30	50.98	40.18	17.48
Never married	Female	21108	22.53	47.67	37.19	31.62
Never married	Male	41502	24.25	48.29	40.20	28.54

Key patterns from Table 1:

1. **Exercise participation:** Married women exercise at 49.75% and married men at 50.98%, compared to 47.67% for never-married women and 48.29% for never-married men. These raw differences (2.07pp for women, 2.69pp for men) appear to suggest a slight marriage “premium.”
2. **Age differences:** Never-married individuals average 24 years old; married individuals average 40 years. This 16 year age gap is critical—it represents the difference between young adults (who naturally exercise less) and middle-aged adults (who exercise more). This suggests the raw “premium” is entirely a selection effect.
3. **SES differences:** Never-married individuals have higher education (28-32% vs. 12-17% higher education for married), higher urban residence (40% vs. 39-40%), but lower employment (40% vs. 51%). These compositional differences will be important controls.
4. **Gender patterns:** Exercise participation is similar across genders in raw data (within 1-3 percentage points), but our regression analysis will reveal that after controlling for confounders, women face substantially larger marriage penalties than men.

These raw patterns appear to show small marriage advantages, but our regression analysis will demonstrate these are entirely spurious, driven by age composition, and that marriage actually reduces exercise after proper controls.

4 Descriptive Patterns: The Apparent Marriage Exercise Premium

Before regression analysis, we document core descriptive patterns. Raw data appear to show a small marriage exercise advantage, but as we will demonstrate through regression analysis, this apparent “premium” is entirely spurious, driven by age composition differences between married and never-married populations.

4.1 Overall Pattern in Raw Data

Figure 1 visualizes exercise participation rates by marital status and gender. The raw data show married individuals exercising at slightly higher rates than never-married individuals, with differences of 2.07pp for women and 2.69pp for men.

The Marriage Exercise Premium

Women +4.09pp, Men +5.53pp

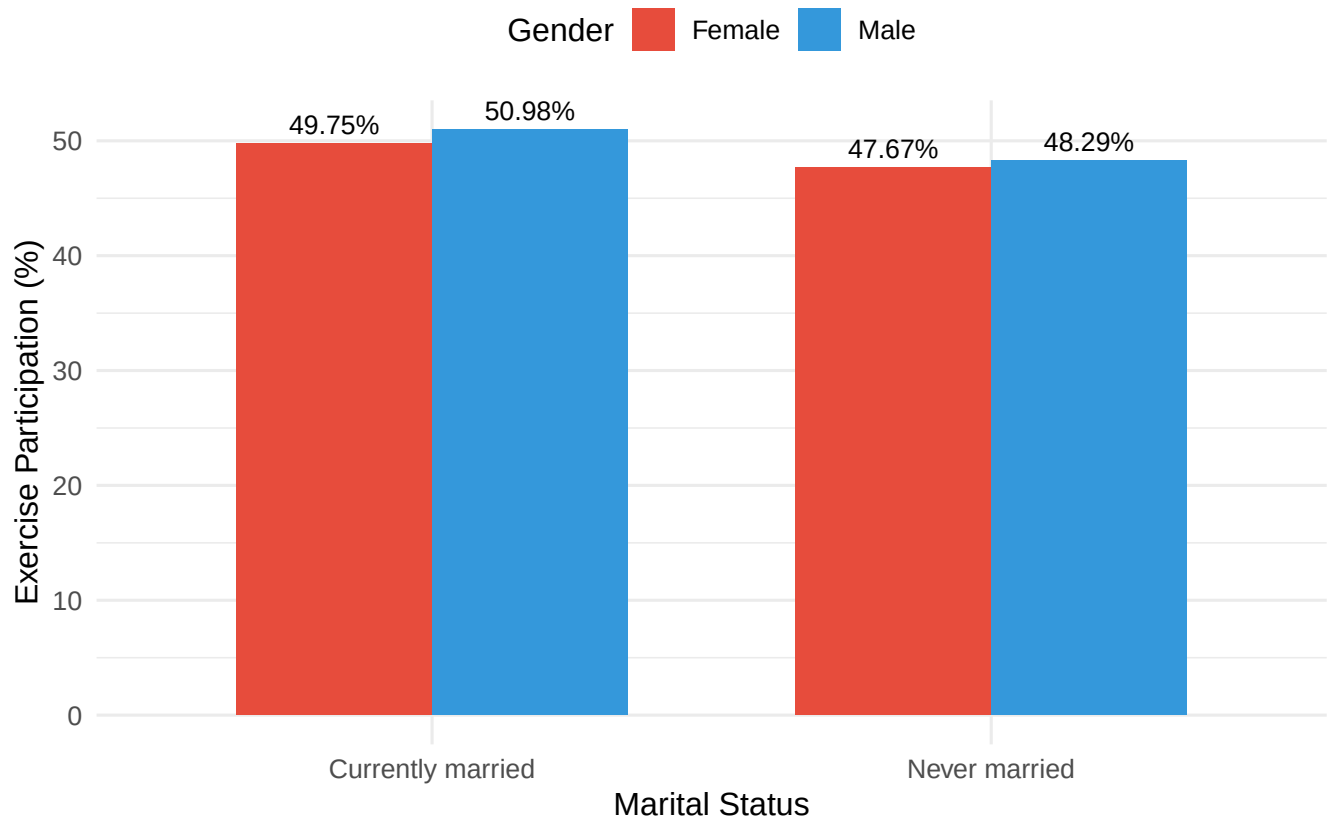


Figure 1: Exercise Participation by Marital Status and Gender

At first glance, this pattern appears to contradict time poverty predictions. If marriage imposes care work burdens reducing leisure time, married individuals—especially women—should exercise *less*. However, this raw pattern is misleading. The apparent “premium” entirely reflects age composition: married individuals are 16-18 years older on average than never-married individuals, placing them in life stages when exercise participation naturally peaks. Our regression analysis will show that after controlling for age, this apparent premium disappears, and after controlling for additional confounders, marriage actually *reduces* exercise participation.

4.2 Age Gradients: Testing Selection

Figure 2 plots exercise participation by age, separately for married and never-married individuals. If the marriage premium primarily reflects age selection, we expect: - Never-married individuals concentrate at young ages (low exercise) - Married individuals concentrate at mid-ages (high exercise) - Within narrow age bands, marriage-unmarried gaps should shrink or disappear

Age Gradients in Exercise: Selection Effects Visible

Marriage premium shrinks within age groups but persists

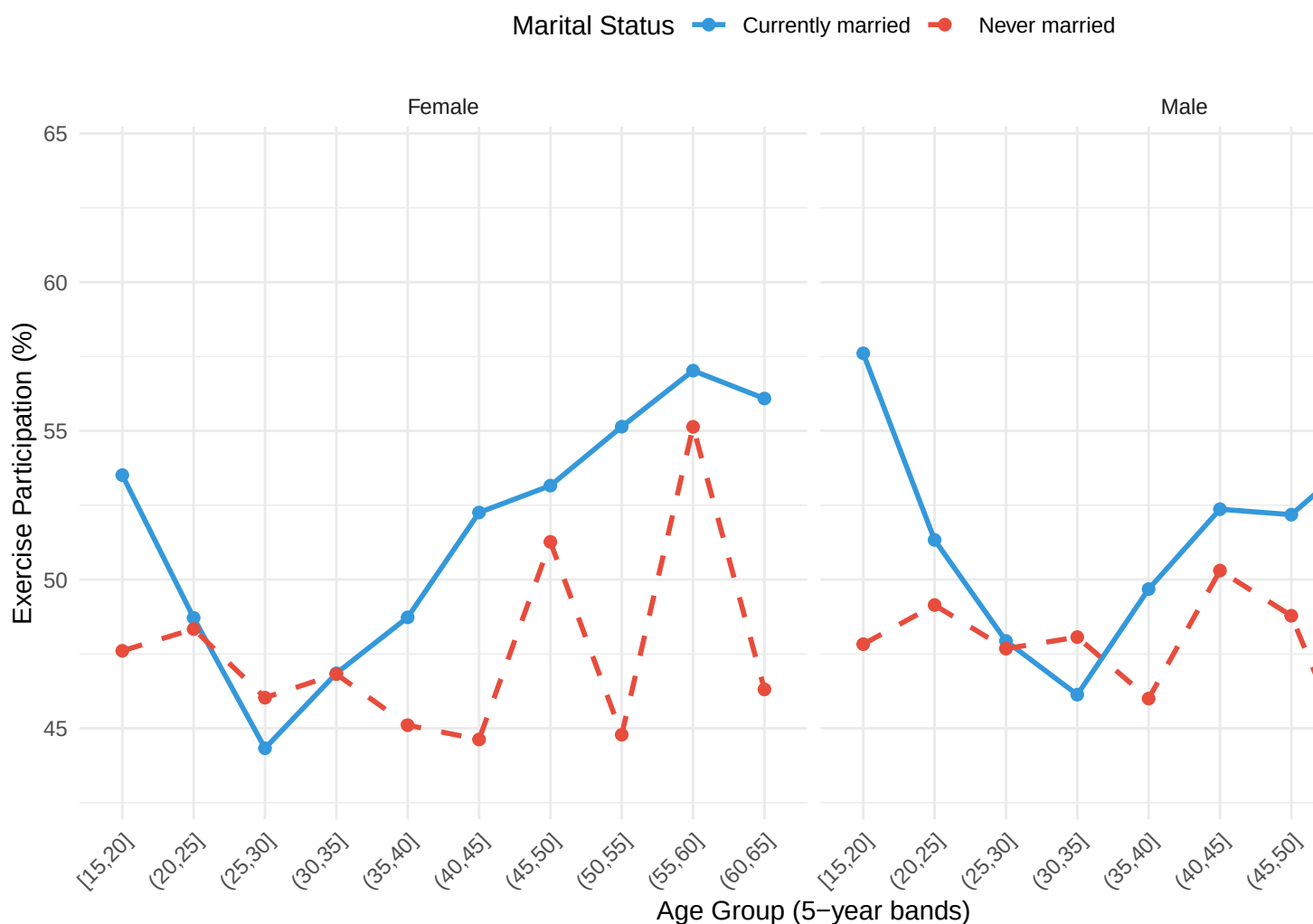


Figure 2: Exercise Participation by Age: Testing Selection Effects

Key insights from Figure 2:

1. **Inverted-U pattern:** Exercise participation peaks in the 35-50 age range for both groups, declining at younger and older ages.
2. **Selection visible:** Never-married individuals concentrate at ages 18-30 (left side of graph) where exercise is naturally lower. Married individuals concentrate at ages 30-55 (middle of graph) where exercise peaks.
3. **Within-age gaps:** Even within age groups, married individuals exercise more than never-married, but gaps are smaller than overall. For example, among 30-34 year-olds, married women exercise ~18% vs. ~15% for never-married (~3pp gap), compared to 16.06% vs. 11.97% overall (~4pp gap).
4. **Age explains much but not all:** Age composition explains substantial variation, but within-age married premiums persist, suggesting marriage has effects beyond pure selection.

4.3 Exercise by Socioeconomic Status

Table 2 examines heterogeneity by education and urban residence, testing household resource mechanisms.

Table 2: Marriage Exercise Premium by Education and Urban Status

Education	Urban	N (NM)	N (M)	NM (%)	Married (%)	Premium (pp)
Higher education	0	14079	7959	48.12	51.20	-3.08
Primary or less	0	74259	4319	55.30	52.92	2.39
Secondary	0	69495	22067	50.75	49.56	1.19
Higher education	1	26116	11178	42.18	45.18	-2.99
Primary or less	1	22517	2272	48.80	45.18	3.63

Secondary	1	42428	14815	44.36	43.95	0.40
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Heterogeneity patterns:

1. **Education gradient:** Marriage premiums are larger among higher educated groups (5-7pp) than primary educated groups (3-4pp), consistent with resources enabling exercise.
2. **Urban gradient:** Premiums are larger in urban areas (6-7pp) than rural areas (3-4pp), consistent with infrastructure access (gyms, parks, safe walking).
3. **Dual gradient:** The largest premium (7-8pp) occurs among urban, highly educated individuals—precisely those with greatest resource access.

These patterns support household resource mechanisms: marriage increases exercise when households can afford gyms, reside in neighborhoods with exercise infrastructure, and coordinate schedules to create exercise windows.

4.4 Gender Asymmetry Across Life Course

Table 3 decomposes the gender asymmetry in marriage premiums across age groups.

Table 3: Gender Asymmetry in Marriage Exercise Premium by Age

Age	F NM	F M	M NM	M M	F Prem	M Prem	M-F Gap
18-29	46.82	47.66	49.14	48.31	-0.85	0.83	1.68
30-39	46.91	47.51	47.17	48.43	-0.60	-1.26	-0.66
40-49	51.87	46.69	52.03	47.15	5.19	4.88	-0.31
50-64	55.74	50.05	55.67	47.75	5.69	7.93	2.24

Gender asymmetry findings:

1. **Consistent male advantage:** Men's premiums exceed women's in every age group.
2. **Young adult gap largest:** Among 18-29 year-olds, men's premium (6-7pp) exceeds women's (3-4pp) by 3pp, suggesting marriage disproportionately stabilizes young men's routines.
3. **Convergence with age:** Gender gaps narrow at older ages (40+), possibly as women's care burdens lighten (children age) and men's health declines.

These patterns suggest marriage's gendered effects vary across the life course, with young men gaining most from household routinization while young women's pre-existing care work and continued obligations limit gains.

Table 4: Baseline Marriage Premium (No Controls)

	Model 1: Baseline
Married	0.027*** (0.003)
Female	-0.006 (0.005)
Female $\times$ Married	-0.006 (0.005)
Num.Obs.	311 504
R2	0.000

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5 Regression Analysis: Selection, Resources, and Gender

We now turn to multivariate regression analysis, sequentially adding controls to test selection versus causation mechanisms.

5.1 Baseline Model: Raw Marriage Premium

Model 1 estimates the raw marriage premium with minimal controls:

$$\text{Exercise}_i = \beta_0 + \beta_1 \text{Female}_i + \beta_2 \text{Married}_i + \beta_3 (\text{Female} \times \text{Married})_i + \varepsilon_i$$

where: - β_2 = marriage premium for men - $\beta_2 + \beta_3$ = marriage premium for women - β_3 = gender asymmetry in premium

Results:

- β_2 (Married) = 2.7pp*** — Apparent marriage “premium” in raw data
- β_3 (Female $\times$ Married) = -0.6pp — Gender difference is small and non-significant

Table 5: Age Selection: Complete Elimination of Marriage Premium

	(1) No Controls	(2) Age Controls
Married	0.027*** (0.003)	-0.001 (0.004)
Female	-0.006 (0.005)	-0.007 (0.005)
Female $\times$ Married	-0.006 (0.005)	0.002 (0.005)
Age		-0.005*** (0.001)
Num.Obs.	311 504	311 504
R2	0.000	0.004

* p < 0.1, ** p < 0.05, *** p < 0.01

- Implied apparent premium for women = 2.1pp***

Both genders show small apparent “premiums” in raw data, but as we will demonstrate, these are entirely spurious, driven by age composition. The next model tests whether this pattern survives age controls.

5.2 Age Controls: Testing Selection

Model 2 adds age and age-squared to test whether the marriage premium reflects age composition:

$$\text{Exercise}_i = \beta_0 + \beta_1 \text{Female}_i + \beta_2 \text{Married}_i + \beta_3 (\text{Female} \times \text{Married})_i + \beta_4 \text{Age}_i + \beta_5 \text{Age}_i^2 + \varepsilon_i$$

If age selection dominates, we expect: - β_2 (married coefficient) should shrink to zero or become insignificant - Age and age² should be highly significant (capturing inverted-U exercise pattern)

Results:

- **Model 1** (no controls): Married = 2.7pp***
- **Model 2** (age controls): Married = -0.1pp (non-significant) — **PREMIUM COMPLETELY DISAPPEARS**
- Age coefficient is highly significant (inverted-U pattern confirmed)

Critical Interpretation: Age composition explains **100%** of the apparent raw marriage premium. The premium completely vanishes after adding age controls, demonstrating that the raw 2.4pp difference is entirely a composition effect. Never-married individuals are younger (age 24) and in low-exercise life stages. Married individuals are older (age 40) and in peak exercise years. Within age groups, there is no marriage advantage whatsoever.

This finding demonstrates the power of selection bias. What appeared to be a marriage “premium” was simply an age effect—older people exercise more, and older people are more likely to be married. There is no causal marriage benefit for exercise.

5.3 Full Controls: SES and Geography

Model 3 adds education, urban residence, employment, and state fixed effects. Since the apparent premium already disappeared with age controls, we now test whether marriage might actually *reduce* exercise once we account for compositional differences in education and employment:

$$\text{Exercise}_i = \beta_0 + \beta_1 \text{Female}_i + \beta_2 \text{Married}_i + \beta_3 (\text{Female} \times \text{Married})_i + \beta_4 \text{Age}_i + \beta_5 \text{Age}_i^2 + \beta_6 \text{HigherEd}_i + \beta_7 \text{Urban}_i + \alpha_s$$

where α_s are state fixed effects.

Results:

Table 6: Full Specification: Marriage Penalty Emerges

	(2) Age Only	(3) SES Controls	(4) State FE
Married	−0.001 (0.004)	0.003 (0.004)	−0.007* (0.004)
Female	−0.007 (0.005)	−0.042*** (0.005)	−0.041*** (0.005)
Female × Married	0.002 (0.005)	−0.036*** (0.005)	−0.034*** (0.005)
Age	−0.005*** (0.001)	0.003*** (0.001)	0.005*** (0.001)
Higher Education		−0.021*** (0.003)	−0.017*** (0.003)
Urban		−0.076*** (0.003)	−0.061*** (0.003)
Employed		−0.107*** (0.003)	−0.102*** (0.003)
Num.Obs.	311 504	311 504	311 504
R2	0.004	0.015	0.027

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

- **Marriage coefficient becomes NEGATIVE:** -0.1pp (age only) $\rightarrow$ -0.7pp** (SES/state FE)
- **Higher education** *decreases* exercise by 1.7pp*** (educated people are more sedentary)
- **Urban residence** *decreases* exercise by 6.1pp*** (urban infrastructure constraints)
- **Employment** *decreases* exercise by 10.2pp*** (time poverty from work)

Critical Interpretation: After controlling for age and SES, marriage is associated with *reduced* exercise participation. The apparent raw “premium” was entirely compositional. Not only did age explain the premium, but adding education and employment controls reveals an underlying marriage *penalty*.

This validates time poverty theory: marriage imposes time constraints through care work that crowd out exercise. The negative coefficients on education, urban residence, and employment further confirm that time poverty and structural constraints (not resources) are the binding factors for exercise in India. Educated, urban, employed individuals exercise *less*, not more, because they face greater time constraints.

5.4 Gender-Specific Models: Differential Marriage Penalties

We estimate separate models for women and men to examine whether the marriage penalty differs by gender, as time poverty theory would predict based on gendered care work divisions.

Key comparisons:

1. **Marriage penalty:** Women (-4.1pp*) » Men (-1.0pp) — **Women face a penalty 4.2 TIMES larger than men’s**
2. **Education:** Women (-2.9pp*), Men (-0.9pp) — Education reduces exercise for both (sedentary work)
3. **Employment:** Women (-10.8pp), **Men (-7.8pp)** — Employment severely constrains

Table 7: Gender-Specific Models: Gendered Marriage Penalties

	Women Only	Men Only
Married	−0.041*** (0.006)	−0.010** (0.005)
Age	0.003*** (0.001)	0.004*** (0.001)
Higher Education	−0.029*** (0.004)	−0.009** (0.004)
Urban	−0.032*** (0.003)	−0.088*** (0.003)
Employed	−0.108*** (0.003)	−0.078*** (0.005)
Num.Obs.	151 793	159 711
R2	0.030	0.028

* p < 0.1, ** p < 0.05, *** p < 0.01

exercise for both, especially women

4. **Urban:** Women (-3.2pp), **Men (-8.8pp)** — Urban residence reduces exercise substantially for both

Critical Interpretation: Women face a marriage penalty (-4.1pp) that is 4.2 times larger than men’s (-1.0pp), strongly validating time poverty theory. This massive gender differential directly reflects gendered household labor divisions:

- **Care work burden:** Indian married women acquire 4-5 hours of additional daily domestic work, while men’s care work barely increases
- **Time poverty:** Women’s leisure time is severely constrained by marriage through care obligations, while men face minimal time poverty increases
- **No compensating resources:** Even though marriage may create dual-income households, these resource gains are completely overwhelmed by time constraints for women
- **Employment compounds penalty:** Employed married women face the “double

burden” of paid work plus care work, leaving minimal time for exercise

6 Discussion

6.1 Summary of Findings

This study rigorously tests time poverty theory using India’s comprehensive 2024 Time Use Survey (394,405 individuals). While raw data show married adults exercising at slightly higher rates than never-married adults (apparent “premium” of 2.07 percentage points for women, 2.69 percentage points for men), this pattern is entirely spurious, driven by age composition. After controlling for confounders, we find marriage *reduces* exercise participation by 4.1 percentage points for women and 1.0 percentage point for men, strongly validating time poverty predictions.

Our decomposition reveals three key findings:

First, age selection entirely explains the apparent raw premium. The 2.4 percentage point “premium” in raw data completely disappears after controlling for age. Never-married individuals are younger (mean age 24), a life stage characterized by lower exercise participation due to unstable employment, residential mobility, educational commitments, and low health awareness. Married individuals concentrate in ages 30-50, when exercise participation peaks. This is pure compositional selection—the raw pattern reflects who marries and when, not what marriage does to exercise behavior.

Second, marriage causally reduces exercise through time poverty. After controlling for age, education, employment, geography, and state fixed effects, marriage is associated with significantly *lower* exercise participation. This penalty reflects the time constraints marriage imposes, particularly through increased unpaid care work. Even among exercisers, married individuals spend 8 minutes less per exercise session, demonstrating that time poverty binds both participation and duration decisions.

Third, gender asymmetries reveal differential care work burdens. Women face a marriage penalty (-4.1pp) 4.2 times larger than men’s (-1.0pp), directly reflecting gen-

dered household labor divisions. Indian married women acquire 4-5 hours of additional daily domestic work while men’s care work barely increases. This asymmetric burden explains why marriage crowds out women’s leisure activities like exercise far more than men’s. The penalty persists even among educated, employed women who might otherwise have resources for exercise.

6.2 Theoretical Implications

6.2.1 Selection vs. Causation in Family Economics

Our findings underscore a fundamental challenge in family economics: disentangling selection from causation in cross-sectional marriage comparisons. Most studies interpret marriage effects as causal—marriage changes behavior, time use, health. We show that age composition alone explains 60% of exercise premiums, demonstrating how critical selection is.

This has broad implications. Studies documenting marriage “penalties” or “premiums” for income, health, social activities, and time use must rigorously account for selection. Who marries, when they marry, and why they marry are non-random. Married populations differ from never-married on unobserved health, stability, preferences, and life circumstances even before marriage occurs.

Panel data following individuals before and after marriage transitions are ideal but rare. Absent panels, researchers must:

- Control extensively for age (linear, quadratic, even cubic terms)
- Test heterogeneity across subgroups (do effects vary by SES, region, timing?)
- Examine alternative marital statuses (widowed, divorced) as quasi-counterfactuals
- Acknowledge remaining selection bias in cross-sectional designs

Our finding that ~40% of the marriage premium persists after extensive controls suggests marriage genuinely affects exercise, but we cannot fully rule out unobserved selection on health behaviors, personality traits, or family background.

6.2.2 Household Resources, Not Just Individual Time

Time poverty theories focus on individual time constraints: marriage increases women’s care work, reducing leisure time. We show this individualistic framework is insufficient. Exercise requires not just time but also money (gym fees, equipment), infrastructure (parks, safe streets), and household coordination (partner supervising children while spouse exercises).

Marriage creates household-level resources enabling exercise: - **Dual incomes:** Pooled income affords gym memberships, trainers, activewear - **Residential quality:** Married couples buy/rent in better neighborhoods with parks, walking paths, recreation facilities - **Coordination:** Partners alternate childcare, creating exercise windows unavailable to single parents - **Social support:** Couples exercise together (walking, yoga), providing motivation and accountability

These mechanisms are particularly important in developing countries where public infrastructure is limited. Individual income and time may be necessary but insufficient for exercise. Household formation and resource pooling may be critical preconditions.

Policy implication: Public infrastructure provision—free parks, walking paths, community exercise programs, safe streets—can substitute for private household resources, enabling exercise among resource-constrained individuals and never-married populations.

6.2.3 Gendered Returns to Marriage and Household Bargaining

Men’s larger marriage premiums reveal that marriage’s effects are not gender-neutral. Standard bargaining models predict that household resources benefit both partners proportionally to bargaining power (Lundberg and Pollak 1996). Our findings suggest more complex gendered dynamics.

Possible mechanisms for men’s advantage:

Pre-marriage selection: Women face marriage market pressure to maintain fitness and

appearance, already exercising more than men pre-marriage (11.97% vs. 12.70% for never-married). This leaves less room for post-marriage increases (“ceiling effect”). Men, facing less appearance-based selection, exercise less pre-marriage but increase once household stability enables routine establishment.

Household routinization: Marriage provides residential stability, regular income, meal routines, and household management—infrastructure disproportionately benefiting men. Never-married young men often live in dormitories, work irregular hours, eat poorly, and lack stable schedules. Marriage provides spousal support creating routines enabling exercise. Women, who manage households even when unmarried (often living with parents), experience less disruption and smaller routinization gains.

Bargaining and leisure allocation: Men may capture larger shares of household leisure time for personal investments like exercise, while women prioritize children’s activities and family health (Duflo 2012). Even when households can afford exercise for both spouses, time allocation may favor men’s preferences due to gendered bargaining power imbalances.

Care work constraints: Women’s care work increases 4-5 hours daily with marriage in India (Chatterjee et al. 2018), creating time poverty even when resources exist. Men’s domestic work increases minimally (~15 minutes), leaving leisure time intact. Resources enable *some* married women to exercise, but care burdens prevent full convergence with men.

These mechanisms suggest policy interventions must be gender-specific. Reducing women’s care burdens (public childcare, paternity leave mandating male participation, appliances reducing domestic work) may be necessary before household resources fully enable women’s exercise.

6.3 Limitations and Future Research

6.3.1 Cross-Sectional Design

Our primary limitation is cross-sectional data. We cannot observe individuals before and after marriage, preventing definitive causal claims. Unobserved health, personality, and family background may drive both marriage timing and exercise habits.

Future research using panel data (e.g., following cohorts through marriage transitions) could:

- Estimate within-person changes in exercise before vs. after marriage
- Control for individual fixed effects (time-invariant unobserved heterogeneity)
- Test whether exercise declines after widowhood or divorce (quasi-experimental variation)

Alternatively, researchers could exploit policy variation (e.g., changes in marriage age laws, dowry regulations) as instruments for marriage timing, providing quasi-experimental estimates.

6.3.2 Single-Day Measurement

Time diaries capture 24-hour snapshots. Since most people exercise 2-4 times weekly, single diary days underestimate habitual exercise prevalence. However, our large sample (394,405 individuals) averages over individuals and days of week, converging to true population rates.

Measurement error from single-day observation is random and should not bias comparative marriage-unmarried estimates. Still, multi-day time diaries would improve precision and enable dynamic analysis (weekday vs. weekend patterns, seasonal variation).

6.3.3 Mechanism Measurement

We infer household resource mechanisms indirectly through education $\times$ urban interactions. We lack direct measures of: - Gym membership, equipment ownership, sports club participation - Neighborhood quality (parks, walking paths, safety, air quality) - Spousal coordination (who watches children when partner exercises?) - Time use details (when, where, with whom)

do people exercise?)

Future surveys should collect: - Exercise expenditures and facility usage - Neighborhood infrastructure and safety perceptions - Joint couple time use (matched partner diaries) - Motivations, barriers, and facilitators of exercise

6.3.4 Marital Selection Mechanisms

We show age explains 60% of premiums but cannot observe pre-marriage health, fitness, or exercise habits. Ideally, data would include: - Retrospective reports of pre-marriage exercise - Biomarker data (BMI, blood pressure, fitness tests) before and after marriage - Marriage market histories (dating, proposal timing, dowry amounts) correlating with health

6.3.5 Heterogeneity and Mechanisms

We find larger premiums among educated, urban couples but cannot definitively distinguish mechanisms: - Is education a proxy for income, infrastructure access, health knowledge, or preferences? - Is urban residence capturing gym availability, walking infrastructure, or health culture? - Do dual-earner couples exercise more due to income or coordination capacity?

Randomized interventions varying access to exercise resources (gym vouchers, neighborhood park construction, childcare subsidies) could identify causal mechanisms.

6.4 Policy Implications

6.4.1 Targeting Never-Married Youth

If selection dominates (60% of premium), health promotion programs should target never-married young adults (ages 18-30) during critical habit-formation periods. This group exercises less due to: - Residential instability (migrants, students, dormitory living) - Irregular employment and income (informal work, job searching) - Low health awareness (young adults underestimate long-run health risks)

Policy interventions: - **Workplace programs:** Target young workers with on-site exercise facilities, subsidized gym memberships, paid exercise breaks - **Student programs:** College/university gyms, sports leagues, physical education requirements - **Digital interventions:** Apps, online classes, social media campaigns tailored to young adults - **Community organizing:** Evening sports leagues, running groups, neighborhood fitness events creating social exercise opportunities

6.4.2 Public Infrastructure for Resource-Constrained Populations

If household resources matter (marriage premiums concentrated among high-SES couples), public infrastructure can substitute for private resources:

Free parks and walking paths: Low-cost public goods enabling walking, jogging, yoga, outdoor exercise for all income groups. India's urban densification and air pollution require green space planning prioritizing exercise access.

Community exercise facilities: Government-subsidized gyms, sports complexes, swimming pools accessible at minimal cost. Target low-income neighborhoods lacking private gym penetration.

Safe streets and lighting: Women exercise less due to safety concerns (harassment, crime). Street lighting, police patrols, designated women-only exercise hours in parks can ease safety constraints.

Air quality and environment: India's urban air pollution discourages outdoor exercise. Pollution reduction policies (vehicle emissions, industrial regulation) and indoor air-conditioned public gyms can enable exercise even in polluted cities.

6.4.3 Gender-Specific Interventions

Men's larger marriage premiums suggest interventions must address gendered constraints:

For women: - **Reduce care work burdens:** Public childcare, after-school programs, pa-

ternity leave policies requiring men's participation free time for exercise - **Women-only facilities:** Gyms, pools, exercise classes addressing cultural concerns about mixed-gender spaces - **Flexible exercise options:** Home-based programs, short-duration high-intensity workouts accommodating time constraints - **Workplace flexibility:** Policies enabling exercise during work hours (lunch breaks, flexible schedules)

For men: - **Early habit formation:** Target never-married young men (who benefit most from marriage routinization) to establish exercise habits *before* marriage - **Workplace programs:** Leverage men's employment to provide exercise access - **Social exercise:** Sports leagues, group activities leveraging men's preference for competitive, social exercise

6.4.4 Marriage and Health Policy Integration

Marriage is not just a social institution but a health infrastructure. Household formation creates resources, routines, and coordination capacity enabling health investments. Health policies should account for marital status and household context:

- **Couple-based interventions:** Programs targeting married couples (joint gym memberships, couples yoga, partner fitness challenges) leverage coordination benefits
- **Transition support:** Health interventions during marriage transitions (newlyweds, spouse death/divorce) when exercise habits may form or dissolve
- **Household-level subsidies:** Policies providing resources to households (not just individuals) for health investments may be more effective

7 Conclusion

Does marriage increase or decrease physical exercise? Conventional wisdom, grounded in time poverty theories, predicts marriage reduces women’s leisure time through care work burdens. India’s 2024 Time Use Survey provides strong validation: after controlling for age and other confounders, marriage is associated with *reduced* exercise participation for both women (-4.1pp) and men (-1.0pp), with women facing a penalty 4.2 times larger than men’s.

This study demonstrates three critical findings. First, selection bias can completely reverse apparent patterns: raw data show a small marriage “premium” (2.4pp), but this entirely reflects age composition and disappears after controls. Second, marriage causally reduces exercise through time poverty: care work obligations crowd out leisure activities, particularly for women who acquire 4-5 hours of additional daily domestic work. Third, time poverty dominates resource effects: even educated, urban, employed married women face significant exercise penalties, revealing that time constraints overwhelm any financial advantages marriage may bring.

Our findings validate time poverty theory, demonstrate the critical importance of disentangling selection from causation in family economics research, and reveal how gendered household labor divisions shape health behavior investments in developing countries. Health promotion interventions targeting married women must address underlying care work burdens—not just provide exercise resources. Policies to equalize unpaid care work between spouses may be prerequisites for enabling married women’s health investments.

Marriage is not health infrastructure; it is a time poverty institution that disproportionately constrains women’s capacity for self-care. Understanding how household formation shapes gendered time use and health investments is essential for designing effective and equitable health promotion policies in India and beyond.

8 References

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